

HTML FORM

How you collect input from users and send it somewhere for processing.

- It **creates a container** for input elements like text fields, checkboxes, radio buttons, and submit buttons.
- It **defines how and where** the data should be sent when the user submits the form.

```
<form action="/submit" method="POST">
  <label for="name">Name:</label>
  <input type="text" id="name" name="name" required>
  <input type="submit" value="Submit">
</form>
```

1. `<label for="name">Name:</label>`

- The `for="name"` attribute links the label with the input whose `id="name"`.
- Clicking on the label will focus the input box. (Accessibility feature for users and screen readers.)

`name="name"` → key used when sending data to the server

If you enter "Ashu", the form sends:

name=Ashu

```
<input type="text" id="name" name="name" required minlength="3"
maxlength="20" pattern="[A-Za-z]+" title="Only alpha
bets allowed">
```

Semantic clarity: It tells browsers and developers "this is a form."

Built-in behavior: Pressing Enter submits the form, validation happens automatically.

Common HTML Form Element

 Common HTML Form Elements	
Element	Purpose
<code><form></code>	Container for all form controls
<code><input></code>	Versatile input field (type defines its behavior)
<code><label></code>	Describes the input field for accessibility
<code><textarea></code>	Multi-line text input
<code><select></code>	Dropdown menu
<code><option></code>	Individual choices inside <code><select></code>
<code><button></code>	Clickable button (can be submit, reset, or custom)
<code><fieldset></code>	Groups related form elements
<code><legend></code>	Caption for a <code><fieldset></code>
<code><datalist></code>	Provides autocomplete options for <code><input></code>
<code><output></code>	Displays result of a calculation or script

Popular `<input>` Types

Type	Description
<code>text</code>	Single-line text input
<code>password</code>	Masked input for passwords
<code>email</code>	Validates email format
<code>number</code>	Accepts numeric input
<code>tel</code>	Telephone number input
<code>url</code>	Validates URL format
<code>search</code>	Search field with optional styling
<code>checkbox</code>	Allows multiple selections
<code>radio</code>	Allows single selection from a group
<code>file</code>	Uploads files
<code>date</code> , <code>time</code>	Date/time pickers
<code>range</code>	Slider input for numeric range

<code>range</code>	Slider input for numeric range
<code>submit</code>	Submits the form
<code>reset</code>	Resets form fields to default values
<code>hidden</code>	Stores data without displaying it
<code>color</code>	Color picker

Fieldset :

Groups related form elements together. It draws a box around them and signals to both users and screen readers that these inputs belong to the same category.

Legend :

Provides a caption or title for the fieldset . It appears at the top of the box and gives context to the grouped inputs.

Autocomplete : it suggest browser prev. enter data

```
<form autocomplete="on">
  <label for="email">Email:</label>
  <input type="email" id="email" name="email" autocomplete="email">

  <label for="name">Name:</label>
  <input type="text" id="name" name="name" autocomplete="name">

  <input type="submit" value="Submit">
</form>
```

Project Assignment :

1. **User Registration & Login System**
2. **Job Application Form**
3. **Survey or Feedback Form**
4. Newsletter Signup with Preferences
5. **Event Registration Form**
6. Contact Us Page

7. Product Order Form
8. Multi-Step Form with Progress Indicator
9. **Profile Settings Page**